

MAT31-6

Math for Data Practice

Regression, regularization, and model evaluation

The mathematical foundations of the data science workflow, built from scratch.

Albert School — 18 hours

Contents

1	Linear regression from scratch	2
1.1	The least squares problem	2
1.2	Geometric interpretation	3
2	Residual analysis	3
2.1	Coefficient of determination	3
2.2	Residual plots and heteroscedasticity	4
3	Multicollinearity	4
4	Ridge regression	4
4.1	Geometric interpretation	4
4.2	Closed-form solution and effect of λ	5
5	Lasso regression	5
5.1	Geometric interpretation and sparsity	5
6	Comparing L1 and L2 regularization	5
7	Cross-validation	6
7.1	Why training error underestimates test error	6
7.2	k-fold cross-validation	6
7.3	Model selection	6
8	The bias–variance tradeoff	6
9	The full regression pipeline	7

1 Linear regression from scratch

We observe n data points, each with p features. Stack the responses in a vector $y \in \mathbb{R}^n$ and the features in a **design matrix** $X \in M_{n,p}(\mathbb{R})$, whose row i holds the features of observation i (a constant column of ones is included to allow an intercept). The **linear model** posits

$$y = X\beta + \varepsilon,$$

where $\beta \in \mathbb{R}^p$ is the unknown coefficient vector and $\varepsilon \in \mathbb{R}^n$ is a random error vector.

1.1 The least squares problem

The **ordinary least squares** (OLS) estimate minimizes the sum of squared residuals:

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|y - X\beta\|^2.$$

Theorem 1 (Normal equations) Any minimizer $\hat{\beta}$ of $\|y - X\beta\|^2$ satisfies the **normal equations**

$$X^T X \hat{\beta} = X^T y.$$

Proof. Let $f(\beta) = \|y - X\beta\|^2 = (y - X\beta)^T (y - X\beta)$. Expanding,

$$f(\beta) = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta.$$

The gradient is $\nabla f(\beta) = -2X^T y + 2X^T X \beta$. Setting it to zero gives $X^T X \beta = X^T y$. The Hessian $2X^T X$ is positive semi-definite, so the critical point is a minimum. $\square$

Proposition 2 (Pseudo-inverse solution) If the columns of X are linearly independent (so X has **full column rank**), then $X^T X$ is invertible and the solution is unique:

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

The matrix $X^+ = (X^T X)^{-1} X^T$ is the **Moore–Penrose pseudo-inverse** of X .

1.2 Geometric interpretation

The set of vectors $X\beta$, as β ranges over $\mathbb{R}^p$, is exactly the **column space** $\text{Col}(X)$, a subspace of $\mathbb{R}^n$. Minimizing $\|y - X\beta\|$ means finding the point of $\text{Col}(X)$ closest to y . By the best-approximation theorem (MAT31-1), that point is the **orthogonal projection** of y onto $\text{Col}(X)$:

$$\hat{y} = X\hat{\beta} = Py, \quad P = X(X^T X)^{-1} X^T.$$

The matrix P is the projection matrix onto $\text{Col}(X)$; it satisfies $P^T = P$ and $P^2 = P$. Thus linear regression is the orthogonal projection of the response vector onto the column space of the design matrix.

2 Residual analysis

The **residual vector** is $r = y - \hat{y} = y - X\hat{\beta}$.

Proposition 3 (Orthogonality of residuals) The residual vector is orthogonal to the column space of X :

$$X^T r = 0.$$

In particular, if X contains an intercept column of ones, the residuals sum to zero.

Proof. From the normal equations, $X^T(y - X\hat{\beta}) = X^T y - X^T X\hat{\beta} = 0$, which is $X^T r = 0$. $\square$

2.1 Coefficient of determination

Write $|y|$ for the mean of the responses. Define the **total**, **residual**, and **explained** sums of squares:

$$\text{TSS} = \sum_{i=1}^n (y_i - |y|)^2, \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{ESS} = \sum_{i=1}^n (\hat{y}_i - |y|)^2.$$

When the model contains an intercept, $\text{TSS} = \text{ESS} + \text{RSS}$.

Definition 4 (R^2 and adjusted R^2) The **coefficient of determination** is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}},$$

the fraction of the variance of y explained by the model; $R^2 \in [0, 1]$. Since R^2 never decreases when a feature is added, the **adjusted R^2** penalizes the number of features p :

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS} / (n - p)}{\text{TSS} / (n - 1)}.$$

2.2 Residual plots and heteroscedasticity

A **residual plot** graphs the residuals r_i against the fitted values $\hat{y}_i$ (or against a feature). Under the model assumptions the residuals should scatter randomly around zero with constant spread (**homoscedasticity**).

Heteroscedasticity is the situation where the variance of the residuals is not constant across the range of fitted values; in a residual plot it appears as a systematic change in vertical spread (for instance a funnel shape, where the spread widens as $\hat{y}$ grows). A visible curved or structured pattern in the residual plot signals that the linear model is misspecified.

3 Multicollinearity

Multicollinearity is near-linear dependence among the columns of X . It makes $X^T X$ nearly singular and destabilises the coefficient estimates.

Definition 5 (Condition number) For the symmetric positive semi-definite matrix $X^T X$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_p \geq 0$, the **condition number** is

$$\kappa(X^T X) = \frac{\lambda_1}{\lambda_p}.$$

A large condition number indicates near-collinearity.

When λ_p is close to 0, the matrix $X^T X$ is nearly singular, so $(X^T X)^{-1}$ has very large entries. The estimate $\hat{\beta} = (X^T X)^{-1} X^T y$ then reacts violently to small changes in y : the coefficients have large variance, can take implausible magnitudes, and may flip sign.

Definition 6 (Variance inflation factor) For feature j , let R_j^2 be the coefficient of determination of the regression of column j on all the other feature columns. The **variance inflation factor** is

$$\text{VIF}_j = \frac{1}{1 - R_j^2}.$$

It measures how much the variance of $\hat{\beta}_j$ is inflated by collinearity; $\text{VIF}_j = 1$ means feature j is uncorrelated with the others, and large values signal multicollinearity.

4 Ridge regression

Ridge regression adds an L_2 penalty to the least squares objective:

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} (\|y - X\beta\|^2 + \lambda \|\beta\|^2), \quad \lambda \geq 0.$$

4.1 Geometric interpretation

This penalized problem is equivalent to minimizing $\|y - X\beta\|^2$ subject to a constraint $\|\beta\|^2 \leq t$ for some t depending on λ . The constraint region $\{\beta : \|\beta\|^2 \leq t\}$ is a **ball** (a disk in two dimensions): the solution is the point where the elliptical contours of the least-squares objective first meet this ball.

4.2 Closed-form solution and effect of λ

Proposition 7 (Ridge solution) For every $\lambda > 0$,

$$\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y.$$

The matrix $X^T X + \lambda I$ is invertible for any $\lambda > 0$, even when $X^T X$ is singular.

Proof. The objective is $\|y - X\beta\|^2 + \lambda\|\beta\|^2$. Its gradient is $-2X^T y + 2X^T X\beta + 2\lambda\beta$. Setting it to zero gives $(X^T X + \lambda I)\beta = X^T y$. $\square$

If $X^T X$ has eigenvalues $\lambda_1, \dots, \lambda_p$ (with eigenvectors fixed by the spectral theorem, MAT31-1), then $X^T X + \lambda I$ has eigenvalues $\lambda_i + \lambda$. Adding λ shifts every eigenvalue **away from zero**, which cures the near-singularity responsible for multicollinearity. In the basis of eigenvectors, the coefficient along direction i is multiplied by $\lambda_i/(\lambda_i + \lambda) < 1$: ridge **shrinks** the coefficients towards zero, shrinking the low-eigenvalue (least stable) directions most. As $\lambda \rightarrow \infty$, $\hat{\beta}_{\text{ridge}} \rightarrow 0$; as $\lambda \rightarrow 0$, it returns to the OLS estimate.

5 Lasso regression

Lasso regression replaces the L_2 penalty by an L_1 penalty:

$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} \left(\|y - X\beta\|^2 + \lambda \sum_{j=1}^p |\beta_j| \right).$$

5.1 Geometric interpretation and sparsity

In constrained form, lasso minimizes $\|y - X\beta\|^2$ subject to $\sum_j |\beta_j| \leq t$. The constraint region $\{\beta : \sum_j |\beta_j| \leq t\}$ is a **diamond** (a cross-polytope), whose corners lie on the coordinate axes. The solution is where the elliptical contours of the least-squares objective first touch this diamond. Because the diamond has sharp corners on the axes, the contact point frequently occurs at a corner, where one or more coordinates of β are exactly zero. This is the **sparsity mechanism**: lasso sets some coefficients exactly to zero and thereby performs feature selection.

Remark The L_1 penalty $\sum_j |\beta_j|$ is not differentiable at the points where some $\beta_j = 0$. Consequently there is **no closed-form solution** for lasso: the estimate must be computed by an iterative numerical optimization procedure.

6 Comparing L1 and L2 regularization

Both penalties shrink coefficients, but their geometry differs:

- **Ridge (L_2)**: the round ball constraint has no corners, so the solution generically has all coefficients non-zero. Ridge shrinks coefficients smoothly and has a closed form. It is preferable when many features each contribute a little and one expects **no** coefficients to be exactly zero.
- **Lasso (L_1)**: the diamond constraint has corners on the axes, so the solution is **sparse**. Lasso is preferable under the **sparsity assumption** — that only a few features genuinely matter — because it performs automatic feature selection by zeroing the rest.

7 Cross-validation

7.1 Why training error underestimates test error

The estimate $\hat{\beta}$ is chosen to minimize the error on the **training** data, so the **training error** is optimistically biased: it is on average smaller than the **test error**, the error on new data drawn from the same distribution. Evaluating a model on the same data used to fit it therefore overstates its quality, and the gap grows with model complexity. Honest assessment requires data not used in fitting.

7.2 k-fold cross-validation

Definition 8 (k-fold cross-validation) Partition the n observations into k disjoint **folds** of roughly equal size. For each $j = 1, \dots, k$:

1. fit the model on all folds except fold j (the training set);
2. compute the mean squared error on fold j (the validation set).

The **cross-validated MSE** is the average of the k validation errors:

$$\text{CV} = \frac{1}{k} \sum_{j=1}^k \text{MSE}_j.$$

Leave-one-out cross-validation (LOOCV) is the special case $k = n$: each fold is a single observation, so the model is fit n times, each time leaving out one point. It uses the data maximally but is the most expensive.

7.3 Model selection

To select the regularization parameter λ (for ridge or lasso), compute the cross-validated MSE over a grid of candidate values of λ and choose the λ that minimizes it. The same procedure selects among models of different complexity.

8 The bias–variance tradeoff

The expected test error of an estimator at a point decomposes into three terms.

Theorem 9 (Bias–variance decomposition) For a target $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2$, and a prediction $\hat{f}(x)$, the expected squared error decomposes as

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{f}(x))}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible}}.$$

- **Bias** measures the systematic error of the model; it is large when the model is too simple (or, for ridge/lasso, when λ is large and shrinkage is strong) to capture the true relationship.
- **Variance** measures sensitivity to the particular training sample; it is large when the model is too complex (or λ is small).
- The **irreducible error** σ^2 cannot be removed by any model.

Increasing model complexity (or decreasing λ) lowers bias but raises variance; decreasing complexity (or increasing λ) does the reverse. The expected test error is minimized at an intermediate complexity. **Numerically**, sweeping λ (or the model complexity) and plotting the cross-validated

error against it exhibits this U-shaped curve, with the minimum at the optimal tradeoff. The variance term here is exactly the estimator variance studied in estimation theory (MAT31-2).

9 The full regression pipeline

Putting the pieces together, the from-scratch regression workflow on a dataset is:

1. assemble the design matrix X (including an intercept column) and response y , and fit OLS via the normal equations $\hat{\beta} = (X^T X)^{-1} X^T y$;
2. compute residuals $r = y - X\hat{\beta}$, report R^2 and adjusted R^2 , and inspect the residual plot for structure or heteroscedasticity;
3. diagnose multicollinearity through the condition number of $X^T X$ and the variance inflation factors;
4. apply regularization — ridge $(X^T X + \lambda I)^{-1} X^T y$ for smooth shrinkage, or lasso under a sparsity assumption;
5. select λ by minimizing the k -fold cross-validated MSE;
6. interpret the final fitted coefficients in light of the bias–variance tradeoff.